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Creative Stories

A Storytelling Game Fostering Creativity

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Outline

- Problem Definition
  - Design machines fostering non-linear thinking
- Proposed Approach
  - Incorporating the Semantic Lateral Thinking technique in a Storytelling Game
- Showcase: Creative Stories
  - Setup
  - Gameplay
- Automating the Assessment Process
  - Computational Creativity Metrics
- Conclusions & Next Steps

Semantic Lateral Thinking

- Involvement of Computers in the Creative Process
  - Computers as independent creative entities
  - Computers as tutors to creative activities
  - *Cooperation of humans-machines for producing creative ideas*
- ***Semantic Lateral Thinking*** (SLT) is well-suited for establishing a cooperation framework
  - Introduction of a foreign conceptual element for disrupting preconceived notions and habitual thought patterns

SLT Computational Tools

- Set of Computational Tools introducing SLT stimuli in the storytelling process

- Thinking Seed Generator
  - Given an initial phrase the tool returns an alternate phrase with some semantic distance from the initial
- Web Miner
  - Crawls the web to discover the dominant terms found in web pages related to an initial search phrase
- Cloud of Thoughts
  - Provides a summary of a text, discovering the main themes and concepts within it
- Competitive Thinking Spaces
  - Clusters the dominant terms of a text segment in the main topics covered by that segment

Showcase: Creative Stories

- a group-based Storytelling Game

Game Flow

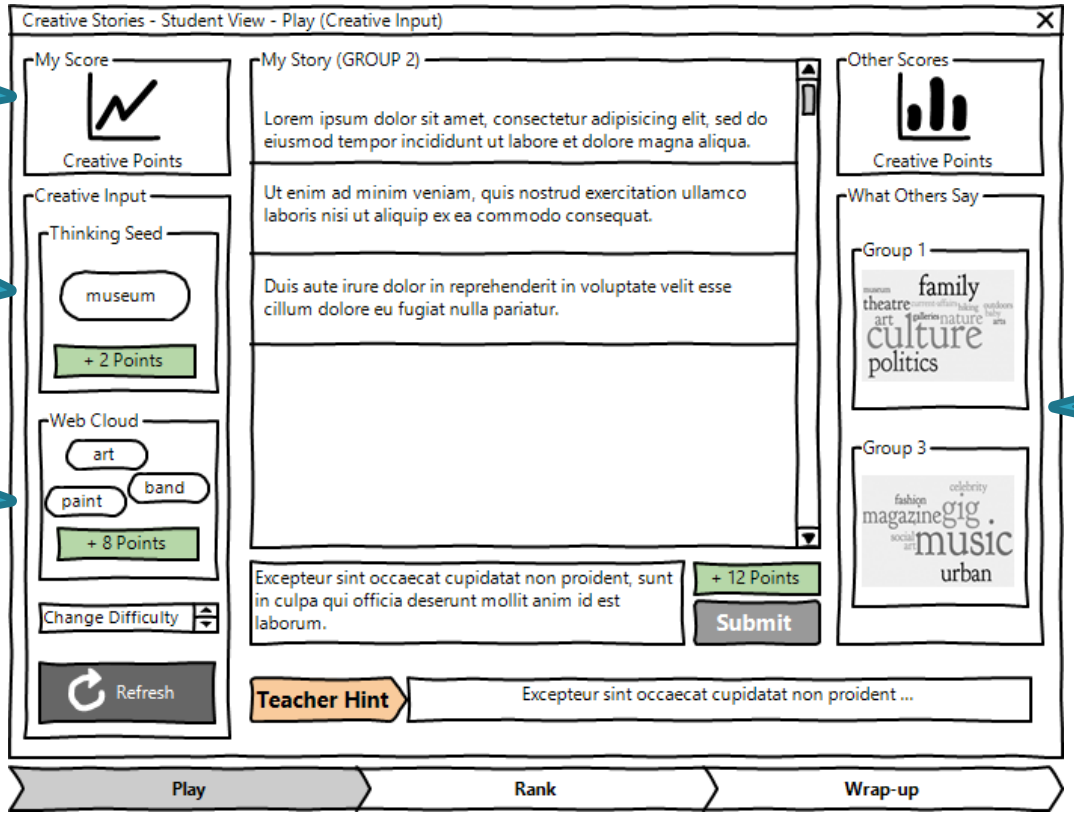
- Players produce their stories in fragments
- They are asked to use SLT stimuli within their stories
  - Creative Input (Thinking Seed + Web Cloud)
  - Competitive Thinking Spaces
- The incorporation of such stimuli increases their game score

Gameplay – Creative Input

Game Score

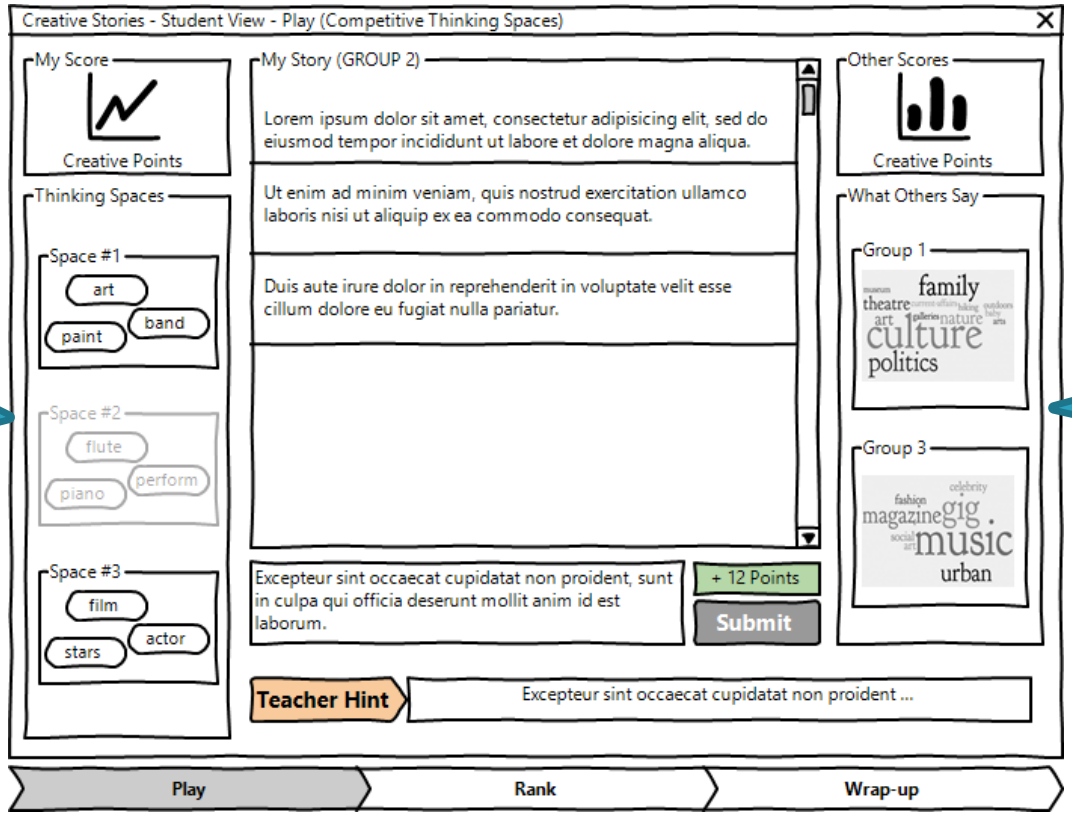
Thinking Seed Generator

Web Miner



Cloud of Thoughts

Gameplay – Thinking Spaces



Competitive Thinking Spaces

Cloud of Thoughts

Automating the Assessment Process

- Scoring in Creative Stories relies on the creativity exhibited during storytelling

- The game incorporates mechanisms for automatically assessing the exhibited creativity
  - Inspired by major computational creativity metrics
  - Formulation of metrics over the specific storytelling context

Computational Creativity Metrics

- Novelty
  - Average semantic distance of dominant terms in the story, compared to the average semantic distance in the whole story set
- Surprise
  - Average semantic distance between consecutive story fragments
- Recreational Effort
  - Sum of weights in the min-weight closure of the story's cluster graph
- Rarity
  - Number of term clusters in the story compared to average number of cluster in the whole story set

Conclusions

- In the context of designing machines fostering non-linear thinking Semantic Lateral Thinking is a well-suited technique
- SLT can be introduced in a creative process via the usage of appropriate computational tools
- In the case of storytelling the SLT disruptors can be lexical entities, semantically and contextually distant from the currently established story
- An automatic creativity assessment method is essential to the process (e.g. via the usage of the proposed metrics)

Next Steps

- Pilot creative stories in a real world setting
- Associate Computational Creativity Metrics with the human perception for Creativity
  - Obtain human rankings for the created stories during the pilots
  - Discover correlation of rankings and computational creativity metrics
- Refine creativity metrics with respect to their assessment over established literary work

Thank You!

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